

HASSAN AHMAD

AI / ML Engineer | Python · Computer Vision · Chatbot Development · API Integration

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PROFILE

Software Engineering graduate with hands-on experience building machine learning, computer vision, and chatbot systems through internships at ITSOLERA, Arch Technologies, and Cornify Technologies. Skilled in Python, classical ML, deep learning, time-series forecasting, and REST API integration. Authored a research paper, "Deep Learning-Based Multimodal Driver Alertness Classification with Missing Modality Robustness," currently ready for IEEE submission. Passionate about developing innovative, ethical AI solutions and delivering real-world impact.

EDUCATION

B.Sc. Software Engineering, CGPA 3.48 / 4.0

Sep 2022 – Jun 2026

University of Malakand, Pakistan

- Field of study: Information and Communication Technologies (EQF Level 6); coursework in machine learning, deep learning, computer vision, data structures, algorithms, and software design
- Final Year Research: "Deep Learning-Based Multimodal Driver Alertness Classification With Missing Modality Robustness" — manuscript ready for IEEE submission

EXPERIENCE

Machine Learning Engineer (Intern)

Oct 2025 – Present

ITSOLERA Pvt. Ltd. · Islamabad, Pakistan

- Built an Image Classification System using traditional ML techniques: extracted HOG features, color histograms, and edge descriptors; trained SVM, KNN, and Random Forest classifiers; evaluated with accuracy, precision, recall, and confusion matrix analysis
- Developed a Time-Series Demand Forecasting System with external variables (promotions, holidays, weather, economic indicators); compared ARIMA, SARIMA, Facebook Prophet, Random Forest, and XGBoost using MAE, RMSE, and MAPE on a rolling-window validation
- Engineered an Adaptive Synthetic Data Augmentation Toolkit: implemented a closed-loop pipeline that performs error analysis on imbalanced cohorts, generates targeted synthetic samples via GANs / VAEs / SDV, and retrain the model iteratively to improve minority-class metrics
- Used Python, scikit-learn, TensorFlow, PyTorch, OpenCV, Pandas, NumPy, Statsmodels, XGBoost, and Matplotlib; managed all code through Git and GitHub

Machine Learning Engineer (Intern, Remote)

Aug 2025 – Sep 2025

Arch Technologies · Pakistan

- Built four end-to-end ML projects: House Price Prediction (regression), Email Spam Classification (NLP text classification), Iris Flower Classification (multi-class), and MNIST Handwritten Digit Recognition (computer vision)
- Handled the full pipeline for each project: data preprocessing, feature engineering, model training, hyperparameter tuning, and evaluation using scikit-learn and TensorFlow

Machine Learning Engineer (Intern, Remote)

Jan 2025 – Mar 2025

Cognifyz Technologies

- Delivered a four-task restaurant analytics project: Restaurant Rating Prediction, Restaurant Recommendation System, Cuisine Classification, and Location-Based Analysis

- Applied regression, classification, recommendation, and geospatial techniques in Python using Pandas, scikit-learn, and Matplotlib; documented findings and model insights for review

SKILLS

AI / ML	scikit-learn, TensorFlow, PyTorch, XGBoost, Statsmodels, Facebook Prophet	Computer Vision	OpenCV, HOG, CNNs, Image Classification, Feature Extraction
Backend & API	Python, FastAPI, REST APIs, Chatbot Development, JSON	Data	Pandas, NumPy, SQL, Time-Series Analysis, Feature Engineering
Generative AI	GANs, VAEs, SDV (Synthetic Data Vault), HuggingFace	Tools & DevOps	Git, GitHub, Jupyter Notebook, VS Code, Docker (basic)
Programming	Python, C++, SQL, Algorithms, Data Structures	Languages	English (C1), Urdu (Native), Pashto (Native)

PUBLICATIONS

Ahmad, H. et al. “Deep Learning-Based Multimodal Driver Alertness Classification With Missing Modality Robustness.” Manuscript ready for submission to IEEE, 2026.

KEY PROJECTS

Multimodal Driver Alertness Classification (Research)

2025 – 2026

University of Malakand — Research Project

- Designed a deep learning system that classifies driver alertness from multiple input modalities (e.g., facial features, eye state, head pose) and remains robust when one or more modalities are missing at inference time
- Built the full pipeline in Python with PyTorch / TensorFlow and OpenCV: data preprocessing, multimodal feature fusion, missing-modality handling strategy, model training, and evaluation
- Authored full research manuscript with introduction, methodology, results, and discussion sections — currently ready for IEEE submission

Restaurant Analytics Suite

Jan 2025 – Mar 2025

Cognifyz Technologies — Internship Project

- End-to-end restaurant data project covering rating prediction, recommendation, cuisine classification, and location-based geographic analysis, presented as a unified deliverable

Computer Vision Mini-Projects

2024 – 2025

Personal & Coursework

- Built MNIST digit recognition with CNNs, image classification with HOG + SVM, and basic object/face detection demos using OpenCV; explored data augmentation strategies and model evaluation

CERTIFICATIONS & MEMBERSHIPS

- Machine Learning Specialization — Stanford University (Coursera)
- Deep Learning for Computer Vision — Online Certification
- IEEE Student Member, University of Malakand Chapter

REFERENCES

Dr. Khalil Ullah
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